

1 System Model and Practical Challenges in Over-the-Air (OTA)

A system that utilizes OTA computing can be modeled in various ways depending on the specific use case and propagation environment. Most architectures involve a set of End Device (ED)s that transmit concurrently to a Fusion Center (FC) or Edge Server (ES). The communication channel is a critical component of this model, as it introduces physical impairments such as fading, noise, and path loss. In addition to the channel model, practical challenges inherent to wireless communications—such as time sampling errors, multipath propagation, and waveform design—can significantly degrade the computational reliability of OTA systems. To mitigate these effects, optimization strategies including power management, transceiver design, and equalization must be employed.

1.a System Architecture and Channel Model

To mathematically formulate the impact of channel impairments on OTA computing, we consider a discrete-time network model consisting of multiple EDs deployed in an agricultural environment. These devices utilize Pulse Amplitude Modulation (PAM) to transmit local temperature readings to a central FC.

Let $x_i[n]$ denote the discrete-time signal transmitted by the i -th sensor. For an illustrative scenario involving two sensors, the transmitted sequences are defined as:

$$x_1[n] = [2, -1, 1] \quad (1)$$

$$x_2[n] = [1, 2, -1] \quad (2)$$

1.a.i The Ideal Target Function

The primary objective of OTA computing is to leverage the multiple access channel (MAC) superposition property to aggregate data natively in the air. Assuming a flat-fading, noise-free, and perfectly synchronized channel, the ideal target function received at the FC, denoted as $s[n]$, is the direct sum of the transmitted signals:

$$s[n] = \sum_{i=1}^K x_i[n] \quad (3)$$

where K is the total number of transmitting devices. For the given sensor data, the ideal aggregated sequence is evaluated as $s[n] = [3, 1, 0]$.

1.a.ii Multipath Channel Distortion

In practical agricultural deployments, the physical environment exhibits multipath characteristics (e.g., reflections from terrain or machinery). This is modeled as a Finite Impulse Response (FIR) filter. Assuming a channel impulse response $h[n]$ with a span of three taps, defined as $h[n] = [1.0, 0.5, -0.2]$, the actual received signal $y[n]$ is the convolution of the ideal aggregated signal with the channel response:

$$y[n] = s[n] * h[n] = \sum_{k=0}^n s[k]h[n-k] \quad (4)$$

Evaluating this convolution yields the distorted sequence $y[n] = [3.0, 2.5, -0.1, -0.2, 0.0]$. The extension of the sequence length from three to five symbols mathematically illustrates the presence of Inter-Symbol Interference (ISI), wherein previous and future symbols bleed into the current sampling instance.

1.a.iii Mean Squared Error (MSE) Evaluation

To quantify the degradation caused by the multipath channel, the Mean Squared Error (MSE) is calculated between the ideal target function $s[n]$ and the actual received signal $y[n]$. The error at each discrete time step is given by $e[n] = y[n] - s[n]$. For a sequence of length N , the overall MSE is defined as:

$$\text{MSE} = \frac{1}{N} \sum_{n=0}^{N-1} (y[n] - s[n])^2 \quad (5)$$

For the evaluated sequences, the discrepancy between the ideal sum and the ISI-corrupted signal results in an $\text{MSE} = 0.46$. This theoretical error establishes the baseline challenge that must be addressed through waveform design or digital equalization techniques at the FC.

1.a.iv Matlab Simulation of Multipath Effects

The following Matlab code snippet simulates the multipath channel effects on the transmitted signals and computes the resulting MSE. The code defines the transmitted sequences, the channel impulse response, and performs the convolution to obtain the received signal. Finally, it calculates the MSE between the ideal and received signals.

Listing 1: Matlab Simulation of Multipath Channel Effects and MSE Calculation

```

1
2      ...
3
4      % Fractional Delay System Object
5      fracDelay = dsp.VariableFractionalDelay('MaximumDelay', 100);
6
7      % Raised Cosine Filter
8      rcFilter = rcosdesign(aleph, span, sps_val, "sqrt");
9
10     % Aggregate noiseless transmitted signal
11     % Initialized as a column vector to prevent MATLAB matrix expansion bugs
12     rx_agg_signal = zeros(N * sps_val + length(rcFilter) - 1, 1);
13
14     for k = 1:K
15         % The Air acts as an ideal adder (MAC Superposition)
16         tx_sig = conv(txUpSamped(:,k), rcFilter(:)) ;
17
18         % Channel: Multipath fractional delay
19         % h(t)

```

In the above code, the ‘`dsp.VariableFractionalDelay`’ object is used to introduce a fractional delay to simulate multipath effects. The ‘`rcosdesign`’ function generates a square-root raised cosine filter to shape the transmitted signals. The received signal is computed by convolving the transmitted signals with the channel response, and the MSE is calculated to quantify the distortion introduced by the multipath channel.

The MSE value differs significantly from the ideal case, highlighting the importance of addressing multipath effects in practical OTA computing systems.

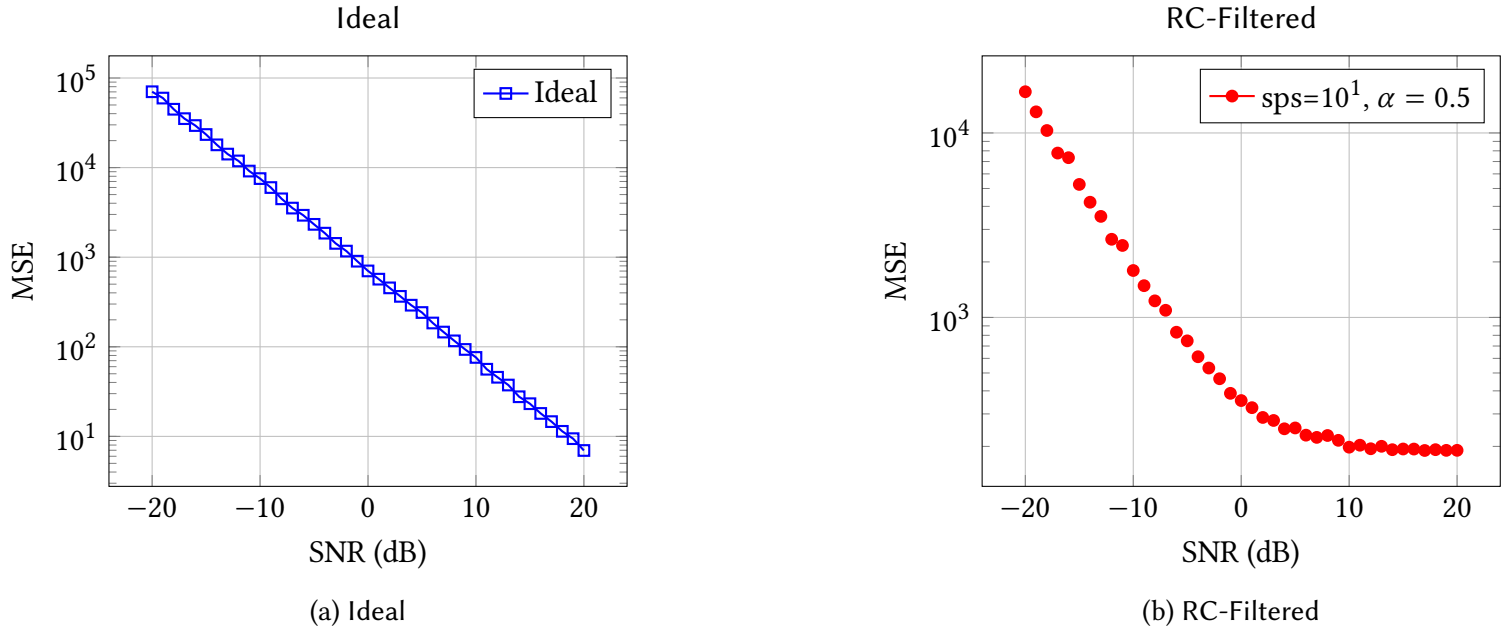


Figure 1: Comparison of MSE performance between the ideal baseline and the Raised Cosine filtered scenario.

1.b Practical Challenges

1.b.i Channel State Information (CSI) Imperfection

The theoretical model assumes perfect knowledge of the channel impulse response. In practice, the need for accurate and fresh Channel State Information (CSI) at transmitters or receivers is critical for precoding and equalization techniques [2], [5], [6]. Obsolete or noisy CSI severely limits the ability to pre-distort signals, exacerbating the MSE.

1.b.ii Waveform and Filter Design

As demonstrated by the sequence expansion in the multipath model, standard digital communication transceivers suffer from time sampling errors and ISI. Designing custom pulse shapes, such as optimized Root Raised Cosine (RRC) filters or DNN-generated waveforms, is essential to force zero-crossings at the exact sampling instances, thereby suppressing ISI natively over the air [2].

1.c Optimization Strategies to Enhance OAC Performance

To mitigate the aforementioned challenges, several optimization frameworks have been proposed in the literature:

- **Power Management:** Strategies to mitigate aggregation errors involve optimizing power control policies at the EDs to align received signal amplitudes, thereby minimizing the overall MSE or maximizing the convergence speed of federated learning algorithms [1], [2], [5].
- **Transceiver Design:** The joint design of pre-processing functions at the transmitter and post-processing equalizers at the receiver allows the system to match the channel characteristics to the desired nomographic function, effectively canceling multipath echoes and minimizing computational distortion [3], [4], [5], [6].

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